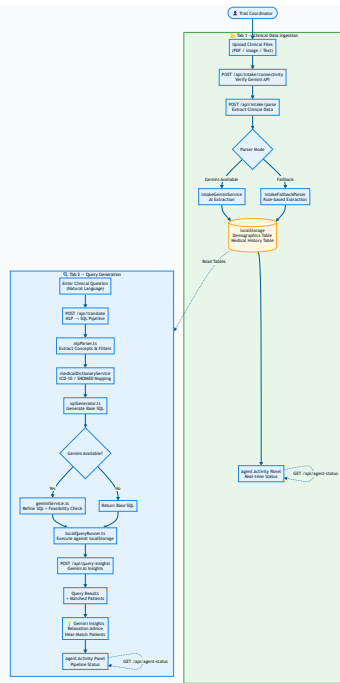


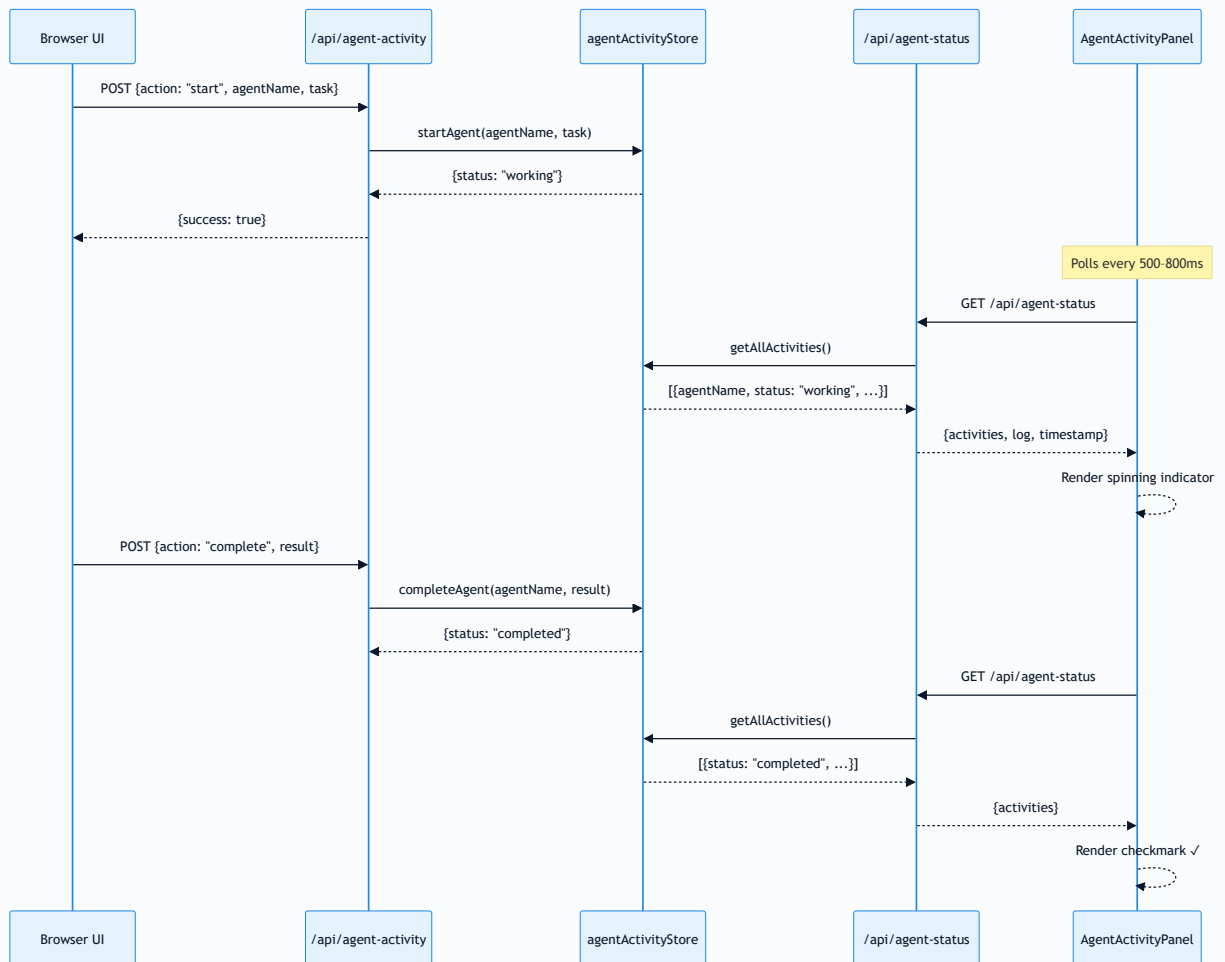
EligibilityAI

Enroll Smarter, Screen Faster — Technical Architecture & Flow Diagram

END-TO-END USER FLOW



AGENT ACTIVITY TRACKING FLOW



TECHNOLOGY STACK

FRONTEND

- Next.js 15.5.3 (App Router)
- React 19.1 + TypeScript
- Tailwind CSS v4
- Lucide React Icons
- Quest DS-Icons (CDN)
- Custom useState / polling hooks

BACKEND / API

- Next.js API Routes (Node.js runtime)
- Google Gemini 1.5 Flash / Pro
- pdf-parse (PDF text extraction)
- FormData API (binary upload)
- agentActivityStore (Map-based)
- localTables (localStorage persistence)

NLP / AI PIPELINE

- Medical Dictionary (ICD-10 / SNOMED)
- Concept extraction (nlpParser.ts)
- Rule-based SQL generation
- Gemini SQL refinement
- Feasibility assessment
- Query insights generation

API ENDPOINTS

Method	Endpoint	Purpose	Key Services
GET	/api/intake/connectivity	Verify Gemini API connectivity before upload	intakeGeminiService
POST	/api/intake/parse	Parse uploaded clinical files into structured records	intakeGeminiService, intakeFallbackParser, pdf-parse
POST	/api/translate	Translate NL question to SQL, run feasibility check	nlpParser, sqlGenerator, geminiService
POST	/api/detect-ambiguities	Detect ambiguous terms in clinical queries	geminiService, medicalDictionaryService
POST	/api/query-insights	Generate AI insights, relaxation advice, near-matches	geminiService, localQueryRunner
POST	/api/agent-activity	Update agent status (start / complete / error)	agentActivityStore
GET	/api/agent-status	Poll current agent activity for real-time display	agentActivityStore

AI AGENT ARCHITECTURE

ORCHESTRATOR

AgentOrchestrator

DATA AGENT

DataIngestionAgent

Coordinates all agents in sequence. Manages pipeline state, routes data between agents, and aggregates results for the UI response.

Handles file parsing and clinical data extraction. Supports PDF, image, and text formats. Falls back to rule-based parsing when Gemini is unavailable.

NLP AGENT

NLPAgent

Extracts medical concepts from natural language queries. Maps terms to ICD-10/SNOMED codes using the medical dictionary service and Gemini disambiguation.

QUERY AGENT

QueryExecutionAgent

Generates and refines SQL queries. Executes against the in-memory patient dataset. Runs feasibility checks and produces structured result sets with near-miss analysis.